

# PlayIT: Offline-First Gamified Early Literacy Mobile Application

## MVP Validation Framework & Instrument Specification (UPA Model — Weeks 1–2 Deliverable)

**Course:** IT411 — Capstone & Research 2 | Semester 1, AY 2026–2027  
**Institution:** College of Computer Studies, Cebu Institute of Technology – University  
**Primary Framework:** UPA (Usability, Pedagogy, Accessibility) | Standard 10-Item SUS + Child Smileometer + Pre/Post Phonics Test  
**Submission Date:** September 13, 2026 | Document Version: 2.0 (Synchronized with Google Form Package)

### 1. Research Rationale & Validation Objective

Early childhood literacy in the Philippines faces critical obstacles, including high student-to-teacher ratios, shortages of individualized instructional materials, and the digital divide characterized by limited or unstable home internet. The Department of Education (DepEd) mandates the **Marungko Approach**—a phono-syllabic reading technique that introduces letters based on phonemic frequency rather than alphabetical order. **PlayIT** serves as a 100% offline-first Android mobile application that integrates the Marungko sequence with Vosk speech recognition, interactive gamification, and diagnostic telemetry.

The Weeks 1–2 MVP Validation is **formative and diagnostic**: (1) evaluate pediatric usability and child task flow; (2) quantify empirical reading gains through structured pre/post assessments; (3) secure expert validation from DepEd reading specialists regarding the 26-letter English Marungko adaptation; and (4) verify offline software stability across target mobile devices.

### 2. Primary Theoretical Model: The UPA Evaluation Framework

For early childhood educational technology, generic UI heuristics are insufficient. PlayIT adopts the **UPA Framework (Usability, Pedagogy, Accessibility)** as its overarching evaluation model, operationalized across four user cohorts:

- **Usability (U):** Evaluates navigation ease, interaction efficiency, and user satisfaction. Operationalized via the **Child Smileometer** (3-point visual scale — Read & MacFarlane, 2006) for young learners, the standardized **10-Item System Usability Scale (SUS — Brooke, 1996)** for parents and teachers, and observational task timings (time-on-task, vocal latency).
- **Pedagogy (P):** Evaluates curriculum alignment, phonetic accuracy, and reading efficacy. Operationalized via the **DepEd Teacher Pedagogy Checklist** (7 Likert items validating Marungko sequence, phoneme modeling, and CVC decodability) and the **Child Pre/Post Phonics Assessment** measuring empirical letter sound and word-blending score gains.
- **Accessibility (A):** Evaluates physical ergonomics and socioeconomic equity. Operationalized via **Pediatric Touch-Target Standards** ( $\geq 64$ dp buttons for motor control), high-contrast visual cues (WCAG 2.1 AA), audio-visual dual-coding, and **100% Offline Zero-Data Operability** with zero ads or tracking.

### 3. Curriculum Scope: 26-Letter Marungko English Adaptation & 33 CVC Blend It Words

PlayIT adapts the 28-letter Filipino Marungko sequence to **26 letters (A–Z)** for Grade 1 English phonics. The Spanish-origin letter **Ñ** does not exist in English orthography. The digraph **NG (ŋ)** is a phonogram rather than an individual grapheme; teaching it as a single letter creates phonological confusion for early readers. The 26 letters are organized into **7 progressive chapters** with **33 decodable CVC Blend It words**:

| Chapter   | Target Letters | Count | Biome Theme        | Canonical Blend It Words (CVC Validated & Asset-Verified) |
|-----------|----------------|-------|--------------------|-----------------------------------------------------------|
| Chapter 1 | m, s, a, i     | 4     | Guava Greenery     | SAM, SIS, AIM (3 words — Group 1 accepted exception)      |
| Chapter 2 | o, b, e, u     | 4     | Mango Orchard      | BUS, SUB, MOM, BEE, BIB                                   |
| Chapter 3 | t, k, l, y     | 4     | Chocolate Hills    | BAT, MAT, KIT, TOY, BOY                                   |
| Chapter 4 | n, g, p        | 3     | Palm Valley        | PIG, PAN, BUG, PIN, NAP                                   |
| Chapter 5 | r, d, h, w     | 4     | Rice Terraces      | DOG, HAT, HEN, BED, WEB                                   |
| Chapter 6 | c, f, j        | 3     | Coral Reef         | CAT, FAN, CAP, CUP, JAM                                   |
| Chapter 7 | q, v, x, z     | 4     | Mount Pulag Summit | VAN, BOX, FOX, ZOO, QUIZ                                  |
| Total     | 26 Letters     | 26    | 7 Biomes           | 33 Decodable CVC Target Words (100% Seeded in Room DB v3) |

#### 4. SMART Evaluation Objectives & Quantitative Targets

| Dimension             | Project Objective                                                    | Evaluation Metric & Instrument                                           | Target Threshold                                                       | Cohort                          |
|-----------------------|----------------------------------------------------------------------|--------------------------------------------------------------------------|------------------------------------------------------------------------|---------------------------------|
| <b>Specific (S)</b>   | Verify child learning & task completion across core sublevels.       | Task completion rate; 1st-pass ASR vocal accuracy; time-on-task.         | >=85% task completion without help; >=75% ASR first-pass accuracy.     | Early Learners (N=12)           |
| <b>Measurable (M)</b> | Quantify usability & caregiver acceptance with standard scales.      | 10-Item SUS score; Child Smileometer; TAM constructs (PU, BI).           | Mean SUS >=75.0 (Grade B+); Smileometer >=2.6/3.0; TAM PU >=4.2/5.0.   | Parents & Teachers (N=15)       |
| <b>Achievable (A)</b> | Validate curriculum alignment & measure empirical learning gain.     | Teacher Pedagogy Checklist (7 items); Child Pre/Post Phonics Assessment. | 100% agreement (>=4.0/5.0); Statistically significant gain (p < 0.05). | DepEd Teachers (N=5) + Learners |
| <b>Relevant (R)</b>   | Ensure 100% offline persistence, data privacy, and diagnostic speed. | Room DB persistence audit; PDF export time; Noise gate test.             | 0% data loss on force-close; PDF export <=3.0s; Noise alert at >40dB.  | Technical Evaluators (N=3)      |
| <b>Time-Bound (T)</b> | Execute validation trials, analyze data, and refactor SRS/SDD.       | Milestone schedule mapped to IT411 Capstone 2 calendar.                  | Protocol: Sept 13; Field Trials: Sept 18; Refactored SRS/SDD: Sept 19. | Research Team (Total N=30)      |

#### 5. Stakeholder Mapping & Questionnaire Item Cross-Walk (UPA Model)

| Code            | Evaluation Criterion / Question Item                                         | UPA Construct              | Target Respondent       | Scale              |
|-----------------|------------------------------------------------------------------------------|----------------------------|-------------------------|--------------------|
| <b>PED-01</b>   | Letter progression strictly follows DepEd Marungko frequency approach.       | Pedagogy: Sequence         | Teachers / SMEs         | 1–5 Likert         |
| <b>PED-02</b>   | Hear It delivers clear, natural phoneme sound modeling.                      | Pedagogy: Audio Quality    | Teachers / SMEs         | 1–5 Likert         |
| <b>PED-03</b>   | Say It uses age-appropriate example words (Mouse, Sun).                      | Pedagogy: Vocalization     | Teachers / SMEs         | 1–5 Likert         |
| <b>PED-04</b>   | Find It uses culturally familiar, unambiguous illustrations.                 | Pedagogy: Visual Discrim.  | Teachers / SMEs         | 1–5 Likert         |
| <b>PED-05</b>   | Blend It uses decodable CVC words respecting cumulative letters.             | Pedagogy: Blending         | Teachers / SMEs         | 1–5 Likert         |
| <b>PED-06</b>   | Excluding NG and Ñ for 26-letter English phonics is pedagogically sound.     | Pedagogy: Scope Validity   | Teachers / SMEs         | 1–5 Likert         |
| <b>PED-07</b>   | Non-punitive gamification (hearts/stars) reinforces mastery resilience.      | Pedagogy: Motivation       | Teachers / SMEs         | 1–5 Likert         |
| <b>PRE-POST</b> | Child phoneme identification and CVC blending diagnostic assessment.         | Pedagogy: Learning Gain    | Early Learners (Tested) | 0–10 Score Delta   |
| <b>SUS 1–10</b> | Standard 10-Item System Usability Scale (Brooke, 1996).                      | Usability: Standard SUS    | Parents & Teachers      | 1–5 Likert (0–100) |
| <b>SMILEY</b>   | Child Smileometer affective rating (Sad=1, Neutral=2, Happy=3).              | Usability: Affective       | Early Learners          | 3-pt Visual Scale  |
| <b>OBS 1–5</b>  | Observational task completion & time-on-task (Map, Hear, Say, Find, Blend).  | Usability: Efficiency      | Child (Observed)        | 3-pt + Seconds     |
| <b>ACC-01</b>   | Pediatric touch targets (>=64dp) prevent miss-taps for small hands.          | Accessibility: Ergonomics  | Parents / Teachers      | 1–5 Likert         |
| <b>ACC-02</b>   | Audio-visual dual-coding enables pre-readers to navigate independently.      | Accessibility: Dual-Coding | Parents / Teachers      | 1–5 Likert         |
| <b>ACC-03</b>   | High-contrast UI colors and large text meet WCAG 2.1 AA standards.           | Accessibility: Contrast    | Parents / Teachers      | 1–5 Likert         |
| <b>ACC-04</b>   | 100% offline access ensures equal opportunity with zero mobile data cost.    | Accessibility: Equity      | Parents / Guardians     | 1–5 Likert         |
| <b>ACC-05</b>   | Ad-free, zero in-app purchase environment guarantees child safety & privacy. | Accessibility: Safety      | Parents / Guardians     | 1–5 Likert         |
| <b>SYS 1–3</b>  | Technical verification: ASR latency <=0.5s, 60 FPS UI, Room DB integrity.    | Technical: ISO 25010       | Technical Evaluators    | Pass / Fail / ms   |

## 6. Detailed Instrument Operationalization (UPA Components)

### Part A: Pedagogy (P) — Pre/Post Phonics Assessment & Teacher Checklist

To establish empirical learning efficacy for Thesis Chapter 4, a 2-minute diagnostic assessment is administered before the child interacts with PlayIT (Pre-Test) and immediately following gameplay (Post-Test):

| Sub-Test                  | Diagnostic Stimulus Items        | Instruction & Administration Protocol                                        | Scoring Rubric                      |
|---------------------------|----------------------------------|------------------------------------------------------------------------------|-------------------------------------|
| Phoneme Sound Recognition | Letter flashcards: M, S, A, I, B | Show card: "What sound does this letter make?" (Prompt /m/, /s/)             | 1 pt per correct sound (Max: 5 pts) |
| Phonetic Picture Match    | Picture cards: Mouse, Sun, Apple | Show 3 images: "Which picture starts with the /m/ sound?"                    | 1 pt per correct match (Max: 3 pts) |
| CVC Word Blending         | Word cards: SAM, BUS             | Show word: "Can you sound out and read this word out loud?"                  | 1 pt per blended word (Max: 2 pts)  |
| Total Diagnostic Score    | 10 Total Phonics Items           | Calculate empirical score gain: Delta = Post-Test Score minus Pre-Test Score | Total Score: 0 to 10 Points         |

### Part B: Usability (U) — Standard 10-Item SUS & Child Smileometer

1. **Child Smileometer (Read & MacFarlane, 2006):** Show the 3 mascot emotion cards immediately post-session: "How did you feel playing with Lily today?" [1 = Sad/Difficult, 2 = Neutral/Okay, 3 = Happy/Fun]. Target: Mean Score  $\geq 2.6$  / 3.0 (Positive Affective Response).
2. **Parent & Teacher System Usability Scale (Brooke, 1996):** 10 standardized items evaluated on a 1–5 scale (Target: SUS  $\geq 75.0$  / Grade B+):
- **SUS-01:** I think that I would like to use PlayIT frequently. | • **SUS-02:** I found the application unnecessarily complex. [R]
  - **SUS-03:** I thought the application was easy to use. | • **SUS-04:** I would need technical support to use this app. [R]
  - **SUS-05:** I found features (Hear, Say, Find, Blend, Dashboard) well integrated. | • **SUS-06:** Too much inconsistency in interface/audio. [R]
  - **SUS-07:** Most children and parents would learn to use this app quickly. | • **SUS-08:** Found the app cumbersome or awkward to use. [R]
  - **SUS-09:** Felt very confident using the app with my child. | • **SUS-10:** Needed to learn a lot of things before getting started. [R]

\*Note: [R] denotes reverse-scored items:  $\text{Score Contribution} = 5 \text{ minus Scale Value}$ . Positive items = Scale Value minus 1. Composite SUS =  $\text{Sum} \times 2.5$  (0–100 scale).

### Part C: Accessibility (A) & Field Protocol

- **Pediatric Ergonomics:** Large touch targets ( $\geq 64\text{dp}$ ) and high-contrast color cards ensure physical ease of use for small child hands.
- **Socioeconomic Equity:** 100% offline operability with zero internet or mobile data requirements guarantees cost-free home practice.
- **Field Administration Protocol:** 5 Android test devices pre-loaded with playit-debug.apk. Ambient noise verified  $\leq 40\text{dB}$ . Each session includes (1) Pre-Test, (2) 15-minute gameplay trial with stopwatch observation, (3) Post-Test, (4) Smileometer rating, and (5) Parent/Teacher SUS survey. Submissions are synced to the master Google Sheet for descriptive and inferential analysis.

## 7. Validation Sign-Off & Thesis Chapter 4 Integration

Empirical findings from this MVP validation will directly support the **IT411 Capstone Midterm Milestone**. Pre/Post score deltas will be evaluated via a **Paired Samples t-test** (or Wilcoxon signed-rank test for non-normal distributions) to establish statistically significant learning gains ( $p < 0.05$ ). System usability scores will be benchmarked using the standard SUS curved grading scale. Insights from teacher qualitative interviews will substantiate the **SRS v3.0 Requirements Traceability Matrix (RTM)** and **SDD v2.0** refactoring due in Week 3.